



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SRI BHARATHI AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

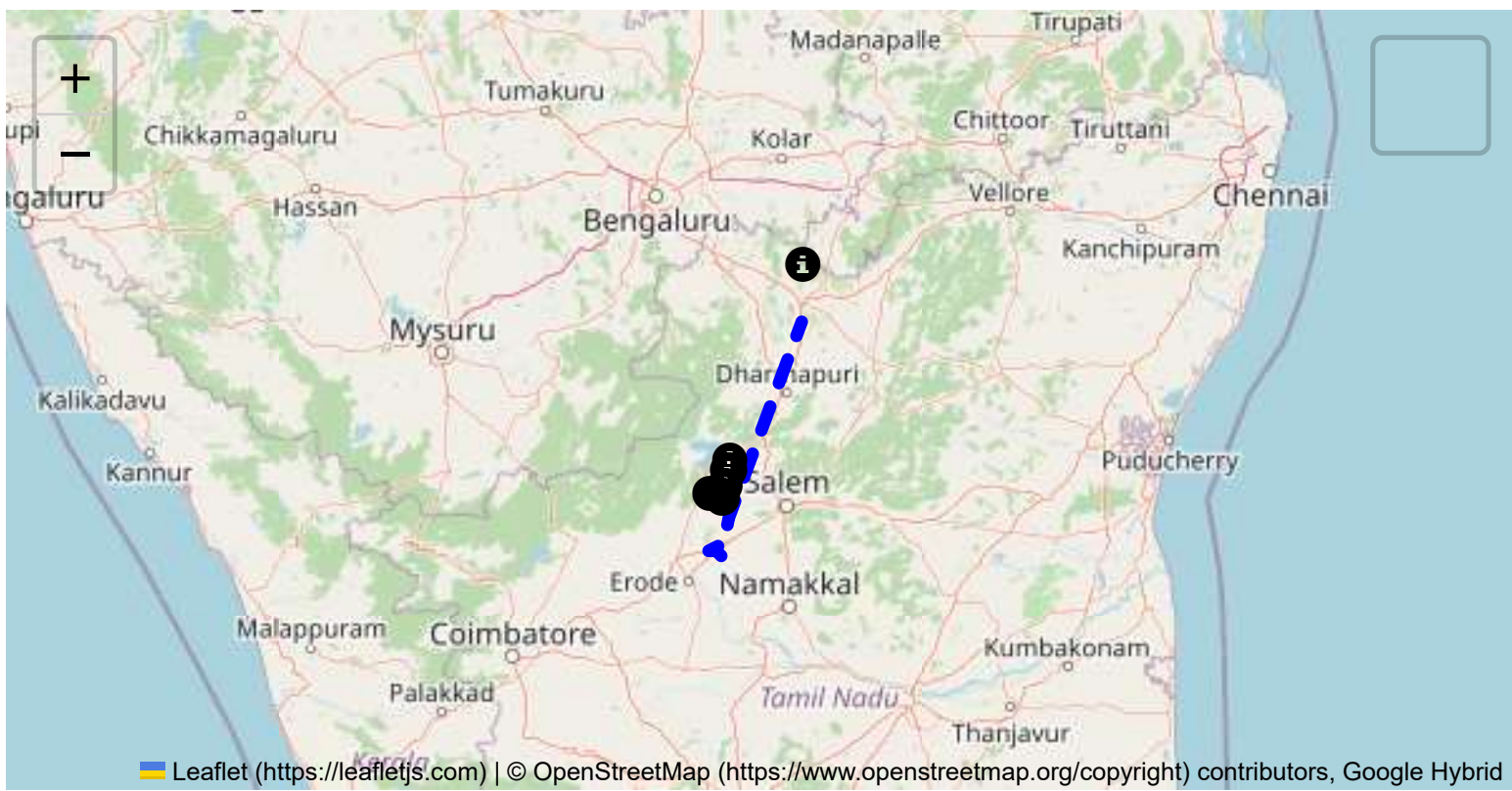
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

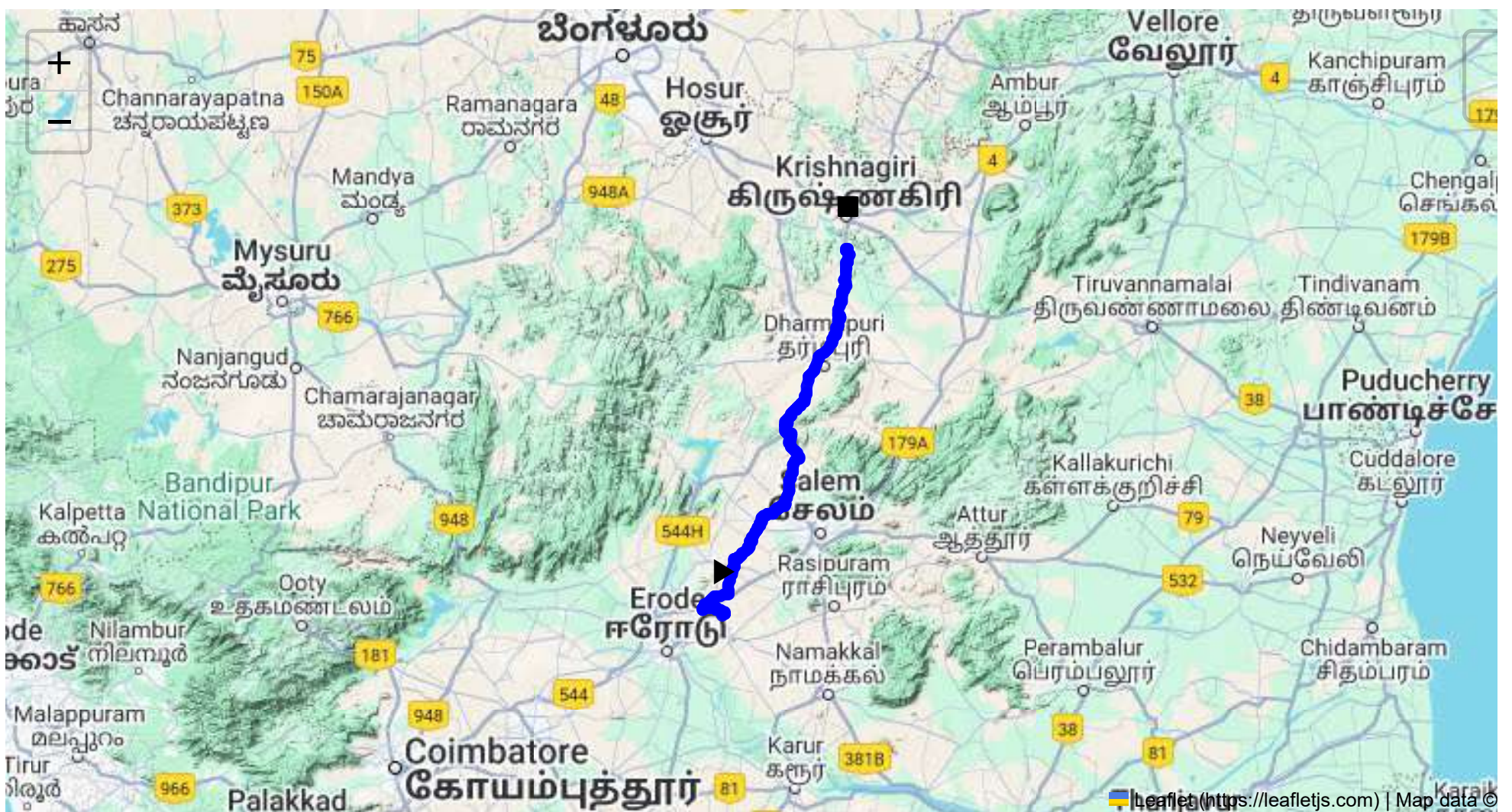
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 142.24 km**  
**Estimated Duration: 2.8 hours**  
**Adjusted Duration (Heavy Vehicle): 3.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.436803, 78.219443)**



## Welcome to the Journey Risk Management Study

- Overview of the Route Map:** The route from CVQF+23W, Sangagiri to 224, Salem Main Rd, Subramaniyapuram, covers approximately 142.24 kilometers through various towns and regions in Tamil Nadu, India. Key areas along the route include Sankari, Bhavani Main Road, and SH 86, passing through several small towns and rural areas. The path involves traveling on connecting roads and the SH 86 highway, providing access to local as well as longer routes.



2. **Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu generally experiences three main seasons: summer (March to June), monsoon (July to October), and winter (November to February). Monsoon season presents the highest risk for weather-related hazards, such as flooding and reduced visibility due to heavy rains. Dry and hot summers could also lead to tire blowouts and heat stress for drivers.
3. **Analysis of Traffic Patterns:** Traffic congestion is likely to be heavier during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) peak hours, particularly near towns and marketplaces. The city centers and main roads are most prone to congestion due to local business activities and transportation of goods. The vicinity of petrol bunks and skill crossings may also experience bottlenecks.
4. **Assessment of Road Quality and Infrastructure:** Road quality varies along the route, with regional roads typically narrower and potentially more worn than the SH 86 highway. Potholes and uneven terrain may be present, especially in less maintained segments near rural areas. Bridges and crossings are generally maintained but should be approached cautiously.
5. **Suggestions for Alternative Routes for Emergencies:** While the primary route is the most direct, traveling through NH 44 and connecting to SH or local roads generally provide alternative paths less influenced by local events. Keeping maps updated with real-time traffic apps can assist in emergency detours.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Transportation of hazardous materials in Tamil Nadu requires adherence to local and national regulations, including obtaining necessary permits, adhering to speed limits, and ensuring proper labeling and containment of materials. Drivers must be trained and certified for handling hazardous goods.
7. **Overview of Historical Incidents:** Specific historical data on incidents involving heavy vehicles might be limited publicly, but general hazards have included road accidents due to over-speeding and poor road conditions, particularly during monsoon. Keeping in contact with local authorities for up-to-date information is critical.
8. **Environmental Considerations and Sensitive Areas:** The route passes through areas of agricultural significance and populated towns, where pollution could have serious impacts. Drivers carrying hazardous materials need to be aware of these areas to avoid spills and ensure substances are securely contained.
9. **Analysis of Communication Coverage:** While most of Tamil Nadu has good telecommunications coverage, there may be sporadic dead zones, particularly in rural or heavily forested regions. Drivers should ensure that communication devices are fully charged, and alternative contact options are available.
10. **Estimated Emergency Response Times:** Urban areas can expect faster emergency response times of 15-30 minutes due to proximity to services. In rural or remote areas, this could extend to 45-60 minutes. Awareness of local emergency service numbers and their stations' locations is essential.
11. **Overall Summary of Risk Assessment:** Overall, the route poses moderate risks due primarily to weather conditions, road quality variances, and traffic congestion during peak hours. Emergency preparedness, careful planning, and adherence to safety protocols are crucial to mitigating these risks

effectively. Enhanced driver training on local regulations and safe material handling can further ensure a safe journey.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45764, 77.85219 | 30 KM/Hr    |
| 5  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.45571, 77.81702 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.45615, 77.81677 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.46916, 77.82431 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.49395, 77.88537 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49465, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.56096, 77.89931 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.57649, 77.90610 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.59214, 77.90656 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.63833, 77.94818 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64171, 77.95184 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.64243, 77.95198 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.64721, 77.95118 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.67870, 77.96846 | 30 KM/Hr    |
| 19 | Turn       | High       | 11.70227, 77.97864 | 15 KM/Hr    |
| 20 | Turn       | Medium     | 11.71715, 77.99969 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.71921, 78.01133 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.72737, 78.02248 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.73775, 78.03465 | 30 KM/Hr    |
| 24 | Turn       | High       | 11.73651, 78.03878 | 15 KM/Hr    |
| 25 | Turn       | Medium     | 11.74189, 78.04660 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.74309, 78.04725 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.77961, 78.05775 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 28 | Turn       | Medium     | 11.86873, 78.08374 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.87966, 78.08187 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.92609, 78.06513 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.93719, 78.05372 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.95828, 78.04983 | 15 KM/Hr    |
| 33 | Turn       | Medium     | 11.95997, 78.04946 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.96335, 78.05141 | 30 KM/Hr    |
| 35 | Turn       | High       | 11.96408, 78.05131 | 15 KM/Hr    |
| 36 | Turn       | Medium     | 11.96511, 78.04966 | 30 KM/Hr    |
| 37 | Turn       | High       | 11.96608, 78.04919 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.96819, 78.05088 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 12.17005, 78.16792 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 12.19919, 78.18172 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 12.22472, 78.19140 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 12.24623, 78.19524 | 30 KM/Hr    |
| 43 | Turn       | Medium     | 12.25078, 78.19988 | 30 KM/Hr    |
| 44 | Turn       | Medium     | 12.25370, 78.20034 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 12.29150, 78.19930 | 30 KM/Hr    |
| 46 | Turn       | Medium     | 12.40604, 78.22285 | 30 KM/Hr    |
| 47 | Turn       | Medium     | 12.42500, 78.22672 | 30 KM/Hr    |
| 48 | Blind Spot | Blind Spot | 12.44082, 78.21942 | 10 KM/Hr    |
| 49 | Turn       | Medium     | 12.43745, 78.21921 | 30 KM/Hr    |

### Emergency Locations

|   | type     | name                       | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------|------------------------|-------------|------------|
| 1 | hospital | Sri Shanthi Hospital       | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2 | hospital | Sugumar Hospital           | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3 | hospital | Omalur Government Hospital | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 4 | clinic   | R.R. Nursing Home          | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 5 | hospital | Sri Hari Hospital          | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 6 | hospital | Sri Sugam Hospital         | 11.7429821, 78.0461039 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 7  | hospital | Saravana Hospital                  | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8  | hospital | Arun Hospital                      | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 9  | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 10 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 12 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 13 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 14 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 15 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 16 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 17 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 18 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 19 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 20 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 21 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 24 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 26 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 27 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 28 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 30 | hospital | Government Hospital                | 12.3048002, 78.2060881 | 30 km/h     | Medium     |

### Crowded Spots

|    | type    | name                                                 | coordinates               | speed_limit | risk_level |
|----|---------|------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                           | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 11 | school  | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 23 | school  | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 25 | college | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |



|    | type    | name                                    | coordinates               | speed_limit | risk_level |
|----|---------|-----------------------------------------|---------------------------|-------------|------------|
| 29 | college | Laxminarayana Polytechnic College       | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |
| 31 | school  | Government Boys Higher secondary school | 12.3047059,<br>78.2065495 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44900, 77.87406





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45764, 77.85219





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45571, 77.81702



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45615, 77.81677





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46916, 77.82431



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.49395, 77.88537





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49465, 77.88488



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56096, 77.89931



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57649, 77.90610





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.59214, 77.90656



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.63833, 77.94818





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64171, 77.95184



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64243, 77.95198



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64721, 77.95118





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.67870, 77.96846



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70227, 77.97864





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71715, 77.99969



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71921, 78.01133



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72737, 78.02248





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73775, 78.03465



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73651, 78.03878





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.74189, 78.04660**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.74309, 78.04725**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.77961, 78.05775**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.86873, 78.08374



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.87966, 78.08187





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92609, 78.06513



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93719, 78.05372



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95828, 78.04983





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95997, 78.04946



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96335, 78.05141





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96408, 78.05131



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96511, 78.04966



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96608, 78.04919





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96819, 78.05088



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.17005, 78.16792





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.19919, 78.18172



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.22472, 78.19140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.24623, 78.19524





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.25078, 78.19988



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.25370, 78.20034





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.29150, 78.19930



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.40604, 78.22285



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.42500, 78.22672





**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 12.44082, 78.21942



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.43745, 78.21921

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